

Minimal Embedding Dimension for Self-Intersection-Free Recurrent Processes on Statistical Manifolds

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Abstract

We study the problem of embedding recurrent processes on statistical manifolds into Euclidean spaces of minimal dimension while preserving temporal distinctness. A *self-intersection* occurs when two temporally distinct points of a trajectory are mapped to the same point in the embedding space. We show that for a class of cyclic processes with strictly monotone meta-time—including recurrent neural dynamics, biological oscillators, and symbolic state transitions—any continuous *phase-preserving* embedding into $\mathbb{R}^2$ (i.e., one that maps the cycle to a simple closed curve, with image depending only on phase and not on meta-time) necessarily produces self-intersections, whereas self-intersection-free embeddings into $\mathbb{R}^3$ always exist. This establishes $k = 3$ as a critical dimension threshold for phase-preserving representations of recurrent processes: $k \leq 2$ forces categorical discretization through unavoidable state conflation, while $k \geq 3$ preserves the continuous structure of temporal dynamics. We interpret this result in terms of metric degeneracy under dimensional collapse: the Fisher information metric loses rank when the meta-time coordinate is projected out.

Keywords: information geometry, dimensional reduction, statistical manifolds, embedding dimension, recurrent dynamics

1 Introduction

Information geometry provides a natural framework for studying inference and decision-making, endowing families of probability distributions with Riemannian structure via the Fisher information metric [1, 2]. In this framework, learning and reasoning correspond to trajectories on statistical manifolds, and the geometry of these manifolds constrains what inferences are possible.

A fundamental question arises when we consider *dimensional collapse*: what happens when a high-dimensional statistical manifold is projected onto a lower-dimensional subspace? This situation is ubiquitous in practice—from rate-distortion theory and the information bottleneck [3] to neural network compression and biological information processing [4].

In this paper, we focus on a specific aspect of dimensional collapse: the emergence of *self-intersections* between temporally distinct states. When a trajectory $\gamma(t)$ on a statistical manifold $\mathcal{M}$ is projected to a lower-dimensional space via a map $\pi_k : \mathcal{M} \rightarrow \mathbb{R}^k$, distinct times $t_1 \neq t_2$ may be mapped to the same point: $\pi_k(\gamma(t_1)) = \pi_k(\gamma(t_2))$. Such self-intersections destroy temporal information and force the system into a discrete, categorical regime.

The question of minimal embedding dimension has a rich history in differential topology, beginning with Whitney’s embedding theorem [5], which establishes that any smooth n -manifold can be embedded in $\mathbb{R}^{2n}$. For dynamical systems, Takens’ embedding theorem [6] provides conditions under which time-delay embeddings reconstruct attractor geometry. Our contribution is to apply these ideas in the information-geometric setting, asking specifically: what is the minimal dimension required to embed *recurrent processes with monotone meta-time* without self-intersection?

Our main results establish that:

1. For cyclic processes with monotone meta-time, self-intersections are structurally unavoidable in $\mathbb{R}^2$ under *phase-preserving* embeddings—those that map the cycle to a simple closed curve with image depending only on phase, not on meta-time. (Embeddings that encode meta-time in the radial coordinate, such as spirals, can avoid self-intersections but violate this constraint.)
2. Self-intersection-free embeddings into $\mathbb{R}^3$ always exist for these processes.
3. The dimension $k = 3$ is therefore a *critical threshold* for phase-preserving representations: categorical below, continuous above.

This result connects to classical dynamical systems theory: the Poincaré-Bendixson theorem [7] states that continuous flows in $\mathbb{R}^2$ cannot exhibit chaotic behavior—only fixed points and limit cycles are possible. Our work provides an information-geometric analog: $k = 2$ constrains representable dynamics to categorical structures, while $k \geq 3$ permits richer process-oriented dynamics.

2 Preliminaries

2.1 Statistical Manifolds and the Fisher Metric

Let $\mathcal{M} = \{p_\theta : \theta \in \Theta\}$ be a parametric family of probability distributions on a sample space $\mathcal{X}$, where $\Theta \subseteq \mathbb{R}^n$ is an open subset. Under regularity conditions, $\mathcal{M}$ carries the structure of a Riemannian manifold with the Fisher information metric [8, 9]:

$$g_{ij}(\theta) = \mathbb{E}_{p_\theta} \left[\frac{\partial \log p_\theta}{\partial \theta^i} \frac{\partial \log p_\theta}{\partial \theta^j} \right]. \quad (1)$$

The Fisher metric captures the local distinguishability of nearby distributions: the geodesic distance between p_θ and $p_{\theta+d\theta}$ measures how statistically distinguishable they are.

2.2 Dimensional Collapse Maps

Definition 1 (Collapse Map). A dimensional collapse map is a smooth map $\pi_k : \mathcal{M} \rightarrow \mathbb{R}^k$ where $k < \dim(\mathcal{M})$. The accessible information geometry on $\mathbb{R}^k$ is the pushforward of the Fisher metric under π_k .

When $k < \dim(\mathcal{M})$, the accessible metric generically has rank less than $\dim(\mathcal{M})$: some directions in parameter space become indistinguishable after projection. This phenomenon is closely related to singularities in statistical learning theory [10].

2.3 Recurrent Processes and Trajectories

Definition 2 (Recurrent Process). A recurrent process on $\mathcal{M}$ is a continuous curve $\gamma : [0, T] \rightarrow \mathcal{M}$ representing the evolution of a belief state or inference trajectory.

Definition 3 (Cyclic Process with Monotone Meta-Time). A recurrent process γ is cyclic with monotone meta-time if:

1. There exists a “phase coordinate” projection $\theta : \mathcal{M} \rightarrow S^1$ such that $\theta \circ \gamma$ covers the circle $n \geq 2$ times.
2. The process encodes a strictly monotone “meta-time” variable $\tau : [0, T] \rightarrow \mathbb{R}$ with $\tau'(t) > 0$ for all t .

Example 4 (Recurrent Systems). This class includes:

- **Biological oscillators:** Neural limit cycles, circadian rhythms, cardiac pacemakers.
- **Recurrent neural networks:** Hidden state trajectories in RNNs processing sequential data.
- **Logical paradoxes:** The Liar’s paradox (“This statement is false”) oscillates between TRUE and FALSE, with each cycle occurring at a distinct meta-time.

Lemma 5 (Intrinsic Cylinder Structure). A cyclic process with monotone meta-time (Definition 3) carries an intrinsic cylinder structure. Specifically, the map

$$\Phi : [0, T] \rightarrow S^1 \times \mathbb{R}, \quad \Phi(t) = (\theta(\gamma(t)), \tau(t)) \quad (2)$$

embeds the trajectory into the cylinder $S^1 \times \mathbb{R}$, where:

- The S^1 factor encodes the cyclic phase coordinate θ .
- The $\mathbb{R}$ factor encodes the strictly monotone meta-time τ .

Since τ is strictly monotone, Φ is injective. The minimal embedding problem thus reduces to: what is the smallest k such that this cylinder-structured trajectory admits a phase-preserving embedding into $\mathbb{R}^k$ without self-intersection?

Proof. Injectivity of Φ follows from strict monotonicity of τ : if $\Phi(t_1) = \Phi(t_2)$, then $\tau(t_1) = \tau(t_2)$, which implies $t_1 = t_2$ by monotonicity. $\square$

Remark 6 (Topological Note). *The cylinder $S^1 \times \mathbb{R}$ can be embedded in $\mathbb{R}^2$ —for example, as a spiral via $(\theta, \tau) \mapsto (e^\tau \cos \theta, e^\tau \sin \theta)$. The obstruction in Theorem 10 arises not from the cylinder topology alone, but from the phase-preserving constraint (Eq. 5), which requires the image to depend only on phase θ , effectively collapsing the cylinder fibers onto a circle. Spiral embeddings violate this constraint by encoding meta-time in the radial coordinate.*

2.4 Self-Intersections and Injectivity

Definition 7 (Self-Intersection). *Given a collapse map π_k and a trajectory $\gamma : [0, T] \rightarrow \mathcal{M}$, a self-intersection occurs when there exist distinct times $t_1, t_2 \in [0, T]$ with $t_1 \neq t_2$ such that:*

$$\pi_k(\gamma(t_1)) = \pi_k(\gamma(t_2)). \quad (3)$$

Equivalently, the composed map $\pi_k \circ \gamma : [0, T] \rightarrow \mathbb{R}^k$ fails to be injective.

Definition 8 (Self-Intersection-Free Embedding). *A collapse map π_k is self-intersection-free for trajectory γ if $\pi_k \circ \gamma$ is injective, i.e.,*

$$t_1 \neq t_2 \implies \pi_k(\gamma(t_1)) \neq \pi_k(\gamma(t_2)).$$

Remark 9 (Collision Measure). *For numerical analysis, we count the collision set:*

$$\mathcal{C}(\pi_k, \gamma) = \{(t_1, t_2) \in [0, T]^2 : t_1 < t_2 \text{ and } \|\pi_k(\gamma(t_1)) - \pi_k(\gamma(t_2))\| < \varepsilon\} \quad (4)$$

for small $\varepsilon > 0$. The embedding is self-intersection-free if and only if $\mathcal{C} = \emptyset$ in the limit $\varepsilon \rightarrow 0$. In practice, we report $|\mathcal{C}|$ for finite ε as an approximate collision count.

3 Main Results

3.1 The Minimal Embedding Theorem

Our main result characterizes when self-intersection-free embeddings exist:

Theorem 10 (Minimal Embedding Dimension). *Let $\gamma : [0, T] \rightarrow \mathcal{M}$ be a cyclic process with monotone meta-time as in Definition 3, with phase projection $\theta : \mathcal{M} \rightarrow S^1$ and meta-time τ . Then:*

- (i) **(Non-existence in $\mathbb{R}^2$)** *Any continuous phase-preserving map $\pi_2 : \mathcal{M} \rightarrow \mathbb{R}^2$ —meaning there exists a simple closed curve $C \subset \mathbb{R}^2$ and a homeomorphism $f : S^1 \rightarrow C$ such that*

$$\pi_2(\gamma(t)) = f(\theta(\gamma(t))) \quad (5)$$

for all t —necessarily has self-intersections: $\pi_2 \circ \gamma$ is not injective.

- (ii) **(Existence in $\mathbb{R}^3$)** *There exists a continuous $\pi_3 : \mathcal{M} \rightarrow \mathbb{R}^3$ such that $\pi_3 \circ \gamma$ is injective (self-intersection-free).*

Remark 11 (Why Phase-Preservation Matters). *The phase-preserving condition (5) excludes embeddings that use “hidden dimensions” to encode meta-time. For instance, a spiral $(\cos \theta, \sin \theta, r(\tau))$ in $\mathbb{R}^3$ projected to $\mathbb{R}^2$ as $(r \cos \theta, r \sin \theta)$ avoids self-intersections by encoding τ in the radius—but this is not phase-preserving, since the image point depends on τ , not just θ . The theorem says: if your 2D representation genuinely factors through phase alone, collisions are unavoidable.*

Proof of (i). Since $\theta \circ \gamma$ covers S^1 at least $n \geq 2$ times and τ is strictly monotone, there exist distinct times $t_1 \neq t_2$ with $\theta(\gamma(t_1)) = \theta(\gamma(t_2))$ (same phase).

By the phase-preserving condition (5):

$$\pi_2(\gamma(t_1)) = f(\theta(\gamma(t_1))) = f(\theta(\gamma(t_2))) = \pi_2(\gamma(t_2)).$$

Thus $t_1 \neq t_2$ but $\pi_2(\gamma(t_1)) = \pi_2(\gamma(t_2))$, so $\pi_2 \circ \gamma$ is not injective. $\square$

Proof of (ii). Define the *canonical helix embedding* $\pi_3 : \mathcal{M} \rightarrow \mathbb{R}^3$ by:

$$\pi_3(\gamma(t)) = \left(\cos(2\pi\theta(\gamma(t))), \sin(2\pi\theta(\gamma(t))), \frac{\tau(t)}{T} \right). \quad (6)$$

This maps the trajectory to a helix in $\mathbb{R}^3$: the first two coordinates encode the phase on a unit circle, while the third coordinate encodes the (normalized) meta-time.

Injectivity: For any $t_1 \neq t_2$, the strict monotonicity of τ implies $\tau(t_1) \neq \tau(t_2)$, so the third coordinates of $\pi_3(\gamma(t_1))$ and $\pi_3(\gamma(t_2))$ differ. Therefore $\pi_3(\gamma(t_1)) \neq \pi_3(\gamma(t_2))$, and $\pi_3 \circ \gamma$ is injective.

Minimality within the phase-preserving class: By Lemma 5, the trajectory has intrinsic cylinder structure $S^1 \times \mathbb{R}$. A phase-preserving embedding (Eq. 5) constrains the image of the S^1 factor to a simple closed curve C whose points depend only on phase. The full trajectory then lives in a tubular neighbourhood of C , and preserving injectivity requires at least one transverse dimension for the $\mathbb{R}$ factor (meta-time). Thus $k = 3$ is minimal *among phase-preserving embeddings*: two dimensions to accommodate the closed curve, one for the monotone coordinate. (Without phase-preservation, spiral embeddings into $\mathbb{R}^2$ are possible; see Remark after Lemma 5.) $\square$

3.2 Information-Geometric Interpretation

The self-intersection phenomenon has a natural interpretation in terms of metric degeneracy.

Proposition 12 (Fisher Metric Singularity). *Consider a statistical manifold parametrized by $(\theta, \tau) \in S^1 \times \mathbb{R}$, where θ is the phase coordinate and τ is the meta-time. Assume the Fisher information matrix $G(\theta, \tau)$ has full rank 2 (both parameters are locally identifiable).*

Under a projection π_2 that discards the τ coordinate, the induced metric on the image has rank at most 1. Specifically, if we write the Jacobian of the projection as J , the induced metric is $\tilde{G} = J^T G J$, which has a null eigenvector in the τ direction. Consequently, the smallest eigenvalue of the induced metric tensor $\tilde{G}$ vanishes, as illustrated numerically in Figure 2A.

This rank drop corresponds to non-identifiability: distinct values of τ produce identical sufficient statistics after projection. Self-intersections are thus equivalent to singularities in the accessible information geometry—points where the induced metric becomes degenerate.

This connects to the broader theory of singular statistical models [10], where such degeneracies affect learning dynamics and generalization.

Numerical proxy: In Figure 2, we illustrate the rank drop using eigenvalues of the empirical covariance matrix as a proxy for the Fisher metric spectrum. This is appropriate for our toy model where the covariance structure tracks the geometric structure; in general statistical settings, the Fisher metric must be computed from the model likelihood.

Example 13 (Gaussian Family with Cyclic Inference). Consider the 2-parameter Gaussian family $\mathcal{M} = \{N(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}$. The Fisher information metric is:

$$G = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix} \quad (7)$$

in coordinates (μ, σ) .

Now consider a cyclic inference process where an agent alternates between two hypotheses about μ while accumulating evidence (increasing certainty, i.e., decreasing σ). Define:

$$\mu(t) = A \cos(2\pi t/P) \quad (\text{oscillating mean hypothesis}) \quad (8)$$

$$\sigma(t) = \sigma_0 e^{-\lambda t} \quad (\text{decreasing uncertainty}) \quad (9)$$

where P is the oscillation period, σ_0 is initial uncertainty, and $\lambda > 0$ is the learning rate.

This defines a trajectory $\gamma(t) = (\mu(t), \sigma(t))$ on $\mathcal{M}$. The phase coordinate is $\theta(t) = 2\pi t/P \bmod 2\pi$, and the meta-time is $\tau(t) = t$ (or equivalently, $\tau = -\log(\sigma/\sigma_0)/\lambda$).

Projection to $\mathbb{R}^2$: If we project via $\pi_2(\mu, \sigma) = (\mu, 0)$ (discarding σ), the trajectory collapses to $[-A, A]$. Distinct times with the same μ (i.e., same phase θ) become identical, producing self-intersections. The induced metric has a null direction: the projection Jacobian kills the σ (equivalently, τ) direction. Note that this rank drop is not due to the intrinsic Fisher metric becoming singular as $\sigma \rightarrow 0$; it is due to the projection discarding σ entirely, so variations in meta-time become undetectable regardless of σ .

Embedding in $\mathbb{R}^3$: The helix embedding $\pi_3(\gamma(t)) = (\cos \theta(t), \sin \theta(t), \tau(t))$ preserves both the cyclic structure and the meta-time, yielding an injective map.

This example shows concretely how dimensional collapse forces loss of the “when” information (meta-time τ) while preserving only the “what” information (phase θ).

3.3 Generalization to Directed Cycles

The result extends beyond simple oscillations to arbitrary recurrent structures:

Theorem 14 (General Cycle Theorem). Let $G = (V, E)$ be a directed graph containing a cycle of length ≥ 2 . Let $\gamma : [0, T] \rightarrow \mathcal{M}^{|V|}$ be a process that:

1. Visits vertices in a pattern that traverses the cycle at least twice.
2. Encodes a strictly monotone meta-time τ .

Then the minimal embedding dimension for self-intersection-free representation is $k_{\min}(\gamma) \geq 3$.

Proof. Embed the cycle $C_n \subset G$ as a simple closed curve in $\mathbb{R}^2$. The visitation pattern induces a phase coordinate $\theta : [0, T] \rightarrow S^1$ by mapping each position along the cycle to a point on the circle.

By hypothesis, $\theta \circ \gamma$ covers S^1 at least twice while τ is strictly monotone. This reduces to the setup of Theorem 10, and the same argument applies.

For graphs with nested cycles (multiple overlapping cycles), each subcycle independently forces $k_{\min} \geq 3$, so the bound holds for the entire graph. $\square$

4 Numerical Illustration

We illustrate the main theorem with explicit numerical examples. Figure 1 shows the self-intersection problem: a cyclic process projected to 2D exhibits self-intersections (marked), while the same process embedded in 3D via a helix is self-intersection-free.

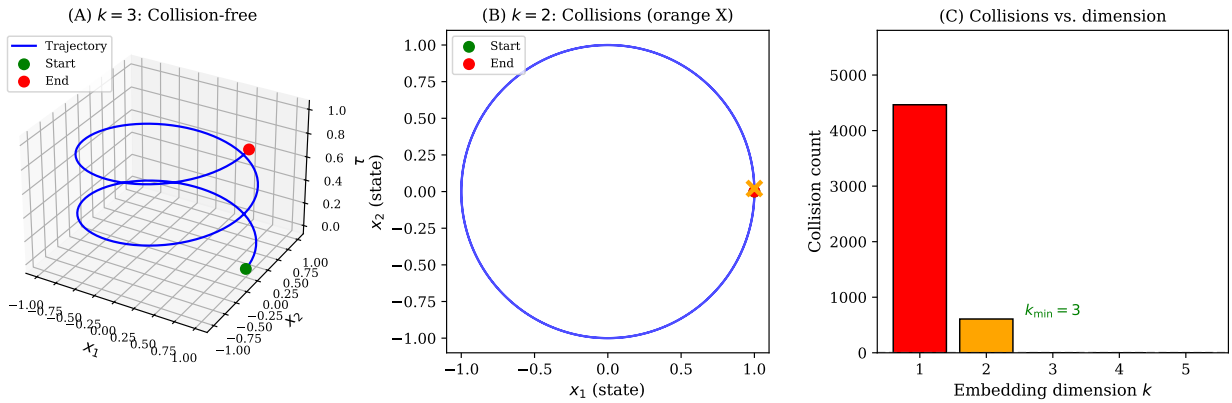


Figure 1: The self-intersection problem in dimensional collapse. (A) A helix in $\mathbb{R}^3$ representing a cyclic process with monotone meta-time—no self-intersections. (B) The same process projected to $\mathbb{R}^2$ —self-intersections occur where the trajectory crosses itself (orange X marks). (C) Self-intersection count as a function of embedding dimension k : self-intersections persist for $k \leq 2$ but vanish for $k \geq 3$. Collision counts depend on the proximity threshold ε ; in the phase-preserving case shown here, collisions persist for all sufficiently small ε once the cycle repeats.

Figure 2 demonstrates the information-geometric interpretation. When the embedding dimension drops below 3, the meta-time parameter becomes non-identifiable, corresponding to a rank drop in the effective Fisher information.

Figure 3 shows that the result generalizes to arbitrary directed cycles, not just simple 2-cycles.

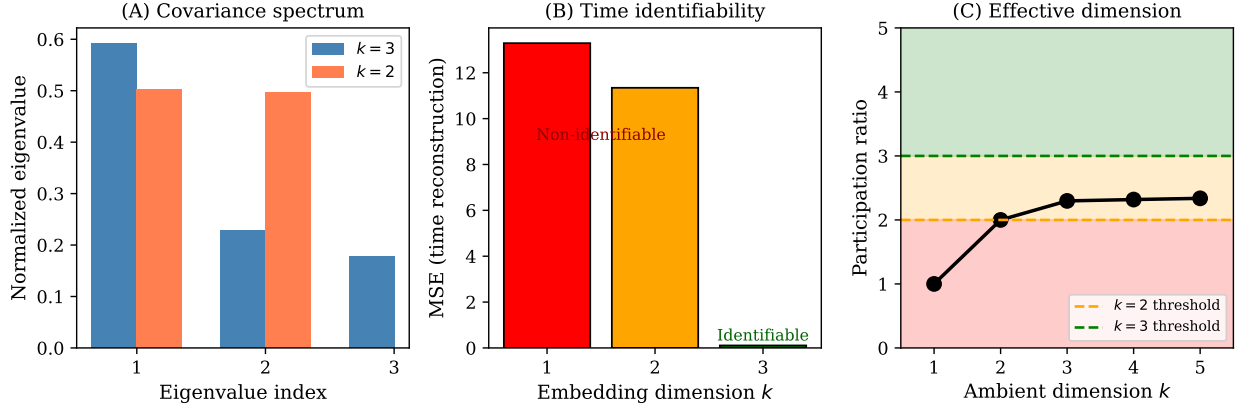


Figure 2: Information geometry of dimensional collapse. (A) Covariance spectrum for $k = 3$ vs $k = 2$ embeddings—the third eigenvalue is lost in projection. (B) Mean squared error in reconstructing meta-time from the embedded coordinates: reconstruction fails for $k \leq 2$ but succeeds for $k = 3$; shaded regions indicate regimes where τ is effectively non-identifiable (red) vs. identifiable (green). (C) Participation ratio (effective dimension) as a function of ambient dimension, showing the critical threshold at $k = 3$.

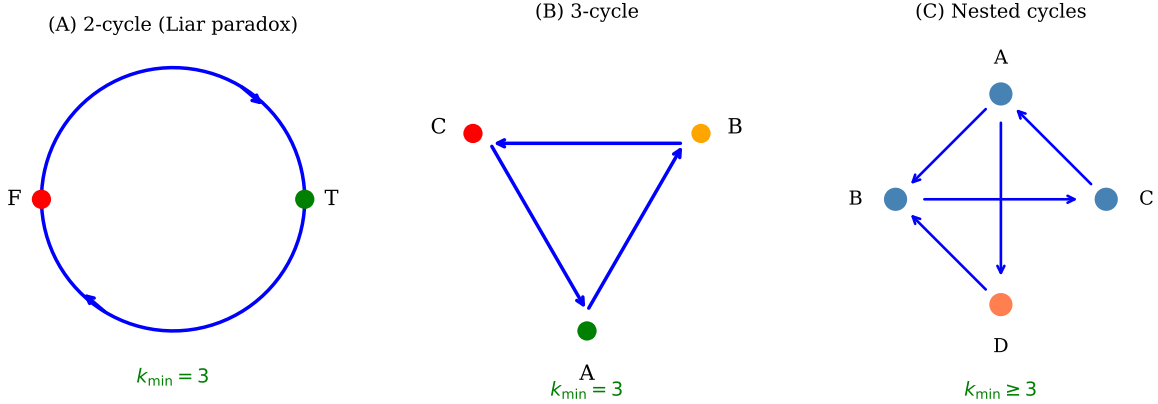


Figure 3: Generalization to directed cycles. (A) A 2-cycle (e.g., TRUE $\leftrightarrow$ FALSE oscillation) requires $k_{\min} = 3$. (B) A 3-cycle $A \rightarrow B \rightarrow C \rightarrow A$ also requires $k_{\min} = 3$. (C) More complex graphs with nested cycles similarly require $k \geq 3$ for self-intersection-free embedding.

5 Discussion

5.1 Categorical vs. Continuous Representations

Our results formalize a fundamental dichotomy in information processing, arising from the tension between *recurrence* (repeating states) and *progress* (monotone meta-time):

- **Categorical regime** ($k \leq 2$): When a recurrent process is constrained to a bounded region in $\mathbb{R}^2$ (a limit cycle rather than an unbounded spiral), self-intersections become unavoidable. The phase-preserving constraint forces dimensional collapse of the trajectory cylinder $S^1 \times \mathbb{R}$ onto a circle S^1 , obliterating the meta-time distinction. The system can only represent “what” state it is in, not “when” it occurred—a geometric bottleneck that naturally engenders discrete, symbolic representations.
- **Continuous regime** ($k \geq 3$): The extra degree of freedom allows the trajectory to spiral helically. The system can maintain strict recurrence in the projected phase plane (x_1, x_2) while encoding progress in the transverse dimension (x_3) . This permits representations of processes that are simultaneously cyclic and temporally distinct.

This dichotomy suggests that the emergence of discrete symbols from continuous dynamics may be geometrically inevitable under dimensional constraints, particularly when the system is spatially confined (as in homeostatic biological systems that cannot drift unboundedly).

Corollary 15 (Discretization as Quotient). *For a cyclic process with monotone meta-time, any $k \leq 2$ embedding that preserves the cyclic structure induces a quotient map on the trajectory. Specifically, let $\sim$ denote the equivalence relation on $[0, T]$ defined by:*

$$t_1 \sim t_2 \iff \pi_k(\gamma(t_1)) = \pi_k(\gamma(t_2)).$$

Then for $k \leq 2$, the equivalence classes $[t] = \{s : s \sim t\}$ are non-trivial (contain more than one element), and the induced map $\bar{\gamma} : [0, T]/\sim \rightarrow \mathbb{R}^k$ is well-defined.

The quotient space $[0, T]/\sim$ is the effective state space of the collapsed representation. For cyclic processes, this quotient is homeomorphic to S^1 : the meta-time information is “quotiented out,” leaving only the phase. Topologically, this corresponds to the bundle projection $\pi : S^1 \times \mathbb{R} \rightarrow S^1$ that fibers the trajectory over the cycle; the “discretization” is precisely the loss of the fiber coordinate. This is the sense in which dimensional collapse forces categorical (discrete-time) structure.

5.2 Connection to Dynamical Systems

Our results resonate with classical results in dynamical systems theory, particularly the Poincaré-Bendixson theorem [7], which states that continuous flows in $\mathbb{R}^2$ cannot exhibit chaotic behavior—only fixed points and limit cycles are possible. In $\mathbb{R}^3$ and higher, strange attractors and complex recurrent behavior become possible.

The parallel is suggestive: $k = 2$ naturally yields simple, categorical structure, while $k \geq 3$ can support richer, process-oriented dynamics. Our contribution is to formalize this in an information-geometric setting.

5.3 Applications

While this paper focuses on the mathematical foundations, the results have potential applications in:

- **Neural coding:** Understanding when neural representations must be discrete vs. continuous [4].
- **Machine learning:** Characterizing bottleneck dimensions in autoencoders and information bottleneck methods.
- **Cognitive science:** Formalizing the emergence of symbolic thought from continuous neural dynamics.

These applications are left for future work.

5.4 Open Questions

Several mathematically precise questions remain:

1. **Piecewise monotone meta-time:** What is $k_{\min}$ if τ is only *piecewise* monotone (allowing reversals)? Conjecture: reversals that return to previously visited meta-times increase $k_{\min}$.
2. **Noisy phase:** If the phase coordinate θ is only *approximately* cyclic (subject to phase slips or noise), does the threshold remain at $k = 3$, or does it depend continuously on the noise magnitude?
3. **Torus dynamics:** For processes on $S^1 \times S^1$ (two coupled phase oscillators with independent monotone meta-times), what is $k_{\min}$? The naive bound is $k_{\min} \leq 5$ (Whitney for 2-manifold), but the actual threshold may be lower.
4. **Non-phase-preserving projections:** If the projection π_k is allowed to *distort* the phase structure (not just collapse the meta-time), can self-intersection-free embeddings exist in $\mathbb{R}^2$? What constraints on π_k are necessary and sufficient?
5. **Collision scaling:** For finite $\varepsilon > 0$, how does $|\mathcal{C}(\pi_2, \gamma)|$ scale with the number of cycles n and the period P ? Is there a universal scaling law?

6 Conclusion

We have established that $k = 3$ is the minimal embedding dimension for self-intersection-free representation of cyclic processes with monotone meta-time on statistical manifolds. This result identifies a critical threshold in information geometry: below three dimensions, temporal information is necessarily lost when the cyclic structure is preserved, forcing categorical representations; at three dimensions and above, continuous processes can be faithfully represented.

The proofs are elementary, relying on the topology of the circle and the definition of injectivity, combined with the classical Whitney embedding principle. Yet the result has implications for understanding when and why discrete structures emerge from continuous substrates.

Statements and Declarations

Funding: No external funding was received for this research.

Competing Interests: The author has no competing interests to declare.

Data Availability: All code for reproducing figures is available at <https://github.com/todd866/minimalembddimension>.

Use of AI Tools: Large language models (Claude 4.5 Opus, Gemini 3 Pro, ChatGPT 5.1 Pro) were used to assist in the drafting of text and the generation of Python plotting code. The author bears full responsibility for the content of the work.

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